

semiautoReg: An R Package for Structure Identification in Partially Linear Regression Models with Autoregressive Errors

Nur Kholifah

*Department of Statistics, Muğla Sıtkı Koçman University
Muğla, Türkiye*

Abstract

This paper introduces semiautoReg, an R package that implements the penalized structure identification procedure of Kazemi et al. (2021) for partially linear regression models with autoregressive errors. In many applications, it is not clear whether a predictor should enter the model as a linear term or as a nonlinear function. This problem becomes more complicated when the data are observed over time and the errors are autocorrelated. The semiautoReg package is designed to handle these issues in one modelling framework. It identifies linear and nonlinear predictors, selects the autoregressive order, and estimates the model parameters using a penalized optimization procedure. The core algorithm combines Group SCAD penalization of centred B-spline coefficient blocks with a scalar SCAD penalty on autoregressive coefficients, minimized via Block Coordinate Descent with Local Quadratic Approximation updates. Tuning is automatic through BIC, AIC, or cross-validation grid search. The package also provides a simple workflow through `semiauto_check()`, `semiauto_prepare()`, `semiauto_fit()`, and `semiauto_plot()`. This package is demonstrated using a monthly Turkish macroeconomic dataset. The results show that semiautoReg can identify interpretable linear and nonlinear effects and provide useful diagnostic plots for model evaluation.

Keywords: *partially linear model, structure identification, SCAD, B-spline, autoregressive errors, R package*

1. Introduction

Regression analysis is the most widely used statistical tool for discovering the relationships between a response variable and a set of predictors. In practice, it is often unclear whether the effect of a predictor should be modeled as linear or as a more flexible nonlinear function. Partially Linear Models (PLMs) provide a useful framework for handling this issue by combining both approaches and partitioning the predictors into a linear set S_1 and a nonparametric set S_2 :

$$y_t = \sum_{j \in S_1} \beta_j x_{tj} + \sum_{j \in S_2} f_j(x_{tj}) + \varepsilon_t, t = 1, \dots, n \quad (1)$$

In many partially linear model methods, the sets S_1 and S_2 are assumed to be known before the model is fitted. However, in real data analysis, this information is usually not clear. If a

nonlinear predictor is treated as linear, the model may give biased results. On the other hand, if too many predictors are treated as nonlinear, the model becomes less simple and more difficult to interpret.

A second issue arises in economic, environmental, and biomedical applications where data are collected over time. In this case, the error terms (ε_t) may be correlated with their previous values. This is known as an autoregressive (AR) process. Ignoring this dependence can reduce the accuracy of the parameter estimates and affect statistical inference.

Kazemi et al. (2021) proposed a penalised identification method to handle these two problems. In this method, all predictors are first treated as nonparametric. The nonlinear parts are then approximated using centred B-spline basis functions. A Group SCAD penalty is applied to the spline coefficient blocks, so nonlinear components that are not needed can be shrunk to zero. In this way, the linear set S_1 can be identified. At the same time, a SCAD penalty is also applied to the AR coefficients to select the relevant lag terms. This procedure is implemented in the R package `semiautoReg`.

At present, there are several R packages related to this type of modelling, but they usually focus on different parts of the problem. For example, `mgcv` (Wood, 2017) and `gamlss` (Stasinopoulos and Rigby, 2007) can fit flexible additive models, but they do not directly identify which predictors should be linear and which should be nonlinear. Penalised regression packages such as `glmnet` (Friedman et al., 2010) and `grpreg` (Breheny and Huang, 2015) are useful for variable selection, but they are mainly designed for parametric regression models. Meanwhile, `PLRModels` (Aneiros and Vieu, 2013) can be used for partially linear regression, but the user still needs to specify the linear and nonparametric parts before fitting the model. To the best of our knowledge, there is currently no publicly available R package that directly implements the structure identification procedure of Kazemi et al. (2021) for partially linear regression models with autoregressive errors. Thus, `semiautoReg` is developed to provide a more automatic workflow for structure identification in partially linear models with autoregressive errors.

This paper is organized as follows. Section 2 explains the statistical model and estimation procedure. Section 3 presents the package architecture. Section 4 illustrates the package using the `macro_turkey` dataset. Section 5 discusses computational details, and Section 6 gives the conclusion.

2. Statistical Methodology

2.1 The Partially Linear Model with AR Errors

Let $\{(y_t, x_t)\}_{t=1}^n$ be a time-ordered sample, where y_t is a scalar response and $x_t = (x_{t1}, \dots, x_{tp})^T$ is a p -dimensional covariate vector. The additive semiparametric model is:

$$y_t = \sum_{j=1}^p f_j(x_{tj}) + \varepsilon_t, t = 1, \dots, n \quad (2)$$

where each $f_j: [0,1] \rightarrow \mathbb{R}$ is an unknown smooth function representing the effect of the j -th predictor. The error ε_t follows a stationary AR(q) process:

$$\varepsilon_t = \sum_{\ell=1}^q \gamma_\ell \varepsilon_{t-\ell} + e_t, e_t \sim iid(0, \sigma^2) \quad (3)$$

The identifiability condition $\mathbb{E}[f_j(x_j)] = 0$ is imposed for all j . Each component function f_j is decomposed as:

$$f_j(x) = \beta_j x + g_j(x) \quad (4)$$

where $\beta_j x$ represents the linear component and $g_j(x)$ represents the nonlinear deviation from linearity. A variable j is assigned to the linear set S_1 if $g_j(x) \equiv 0$, and it is assigned to the nonlinear set S_2 if $g_j(x)$ is not identically zero. Hence, structure identification amounts to determining which nonlinear deviation functions g_j are identically zero.

2.2 B-Spline Approximation

Each unknown nonlinear component g_j is approximated using centered B-spline basis functions. The interval $[0,1]$ is divided by K interior knots, which are selected based on the quantiles of the empirical distribution of x_j . Let $\bar{K} = K + d$, where d is the spline degree (default $d = 3$, cubic). The normalized B-spline basis functions $B_1(x), \dots, B_{\bar{K}}(x)$ are centered as:

$$\tilde{B}_k(x) = B_k(x) - \frac{1}{n} \sum_{i=1}^n B_k(x_i), k = 1, \dots, \bar{K} \quad (5)$$

The nonlinear component g_j is then approximated as $g_j(x) \approx \tilde{B}_j(x)^T \theta_j$, where $\theta_j \in \mathbb{R}^{\bar{K}}$ is the vector of spline coefficients. By combining all linear and nonlinear components, the model can be written in matrix form as:

$$y = X\beta + Z\theta + \varepsilon \quad (6)$$

Here, X is the $n \times (p + 1)$ linear design matrix including the intercept, $\beta \in \mathbb{R}^{p+1}$ is the parametric coefficient vector, $Z = [Z_1, \dots, Z_p]$ is the $n \times p\bar{K}$ spline design matrix, and $\theta = (\theta_1^T, \dots, \theta_p^T)^T$ collects all spline coefficient vectors.

2.3 Profile Least Squares and AR Correction

Let $P_x = X(X^T X)^{-1} X^T$ be the projection onto the column space of X . Define $y^* = (I - P_x)y$ and $Z^* = (I - P_x)Z$. For fixed autoregressive coefficient vector $\gamma = (\gamma_1, \dots, \gamma_q)^T$, the AR-corrected partial residual is:

$$\tilde{y}_t = y_t^* - \sum_{l=1}^q \gamma_l y_{t-l}^*, \tilde{Z}_t = Z_t^* - \sum_{l=1}^q \gamma_l Z_{t-l}^*, t = q + 1, \dots, n \quad (7)$$

The AR-corrected least-squares loss function is:

$$\mathcal{L}(\theta, \gamma) = \sum_{t=q+1}^n (\tilde{y}_t - \tilde{Z}_t^T \theta)^2 \quad (8)$$

2.4 Penalized Objective and SCAD

Structure identification and lag selection are performed by minimizing the following penalized criterion:

$$Q(\theta, \gamma) = \mathcal{L}(\theta, \gamma) + \lambda \sum_{j=1}^p p_{\text{SCAD}}(\|\theta_j\|; \lambda) + \mu \sum_{\ell=1}^q p_{\text{SCAD}}(\|\gamma_\ell\|; \mu), \quad (9)$$

where $\|\cdot\|$ denotes the Euclidean norm. The function p_{SCAD} is the Smoothly Clipped Absolute Deviation penalty proposed by Fan and Li (2001), defined as

$$p_{\text{SCAD}}(t; \lambda) = \begin{cases} \lambda t, & 0 \leq t \leq \lambda, \\ \frac{a\lambda t - (t^2 + \lambda^2)/2}{a-1}, & \lambda < t \leq a\lambda, \\ \frac{(a+1)\lambda^2}{2}, & t > a\lambda, \end{cases} \quad (10)$$

with $a = 3.7$, as recommended by Fan and Li (2001). Because the penalty is applied to the norm $\|\theta_j\|$, the spline coefficients of predictor j are selected as a group. Therefore, the whole nonlinear component is either retained in the model or removed from the model.

2.5 Estimation Algorithm

Since p_{SCAD} is non-smooth, Fan and Li (2001) proposed the Local Quadratic Approximation (LQA), which approximates the penalty locally by a quadratic function at the current iterate $\theta^{(n)}$:

$$p_{\text{SCAD}}(\|\theta_j\|) \approx p_{\text{SCAD}}(\|\theta_j^{(n)}\|) + \frac{1}{2} \left[\frac{p'_{\text{SCAD}}(\|\theta_j^{(n)}\|)}{\|\theta_j^{(n)}\|} \right] (\|\theta_j\|^2 - \|\theta_j^{(n)}\|^2) \quad (11)$$

The complete Block Coordinate Descent (BCD) algorithm is summarized as follows:

1. Initialize $\theta^{(0)}$ using ridge-regularized Ordinary Least Squares (OLS) on the purged system, and $\gamma^{(0)}$ via OLS on initial residuals.
2. Update $\theta^{(n+1)}$ for each group j , construct the corresponding partial residual, compute the unconstrained block OLS estimate u_j , and apply group SCAD thresholding.
3. Update $\gamma^{(n+1)}$, recompute the residuals using the updated value $\theta^{(n+1)}$, and then solve the LQA-penalized normal equations for the autoregressive parameters.
4. Repeat the updates until the convergence criterion $\max(\|\theta^{(n+1)} - \theta^{(n)}\|, \|\gamma^{(n+1)} - \gamma^{(n)}\|) < 10^{-6}$ is satisfied.
5. Apply hard-thresholding after convergence by setting $\theta_{jk} = 0$ if $|\theta_{jk}| < 10^{-8}$, and $\gamma_l = 0$ if $|\gamma_l| < 10^{-6}$.

After convergence, the identified linear and nonlinear sets are defined as $\hat{S}_1 = \{j: \|\hat{\theta}_j\| = 0\}$, $\hat{S}_2 = \{j: \|\hat{\theta}_j\| > 0\}$. The selected active autoregressive lags are given by $\{l: |\hat{\gamma}_l| > 0\}$. Finally, the linear coefficients are recovered as $\hat{\beta} = (X^T X)^{-1} X^T (y - Z\hat{\theta})$.

2.6 Tuning Parameter Selection

The AR order q is selected before the final model fitting. It is chosen by comparing AR(1), ..., AR(q_{max}) models fitted to the initial OLS residuals, and the model with the smallest BIC value is selected. This q -selection step always uses BIC and is independent of the tuning criterion chosen for λ and μ . In this package, the default maximum order is $q_{\text{max}} = 5$. The penalty parameters λ and μ are then selected jointly using a two-dimensional grid search. Three tuning criteria are available in `semiauto_fit()`, as shown in Table 1.

Table 1. Tuning parameter selection methods in `semiauto_fit()`

Method	Speed	Description
BIC	Fast	Bayesian Information Criterion. This is the default option and usually gives a sparser model.
AIC	Fast	Akaike Information Criterion. It may select a slightly less sparse model than BIC.
CV	Slow	5-fold cross-validation. It is useful for prediction, but not recommended for large grids.

3. Package Design

3.1 Installation

The package can be installed from CRAN once it is released. During development, it can be installed directly from GitHub.

```
# From CRAN
install.packages("semiautoReg")

# Development version
remotes::install_github("nkhnurkholifah-maker/semiautoReg")
```

3.2 Package Architecture

The semiautoReg package contains five main functions and one example dataset. These tools help users from data checking to model diagnostics and visualization, covering the whole workflow, as shown in Table 2.

Table 2. Exported functions and dataset in semiautoReg

Function	Class returned	Purpose
semiauto_check()	semiauto_check	Validate input data quality
semiauto_missing()	semiauto_missing	Summarize missing values by variable
semiauto_prepare()	semiauto_prepare	Clean, sort, and build model matrices
semiauto_fit()	semiauto_fit	Fit the PLM-AR model (main function)
semiauto_plot()	list/patchwork	Produce diagnostic plots
macro_turkey	data.frame	Turkish macroeconomic example dataset

S3 generics print, summary, fitted, residuals, and predict are registered for the semiauto_fit class, allowing users to interact with model objects using standard R syntax.

The package workflow begins with semiauto_check(). This function ensures the data are ready for modeling by checking response and predictor variables, spotting missing values, duplicate rows, and constant variables. It also checks the minimum sample size requirements.

After semiauto_check(), semiauto_prepare() prepares the data before building any models. It selects the variables specified by the user, manages missing values through omitting or failing actions, sorts the data by time if a time variable is provided, and constructs the response vector and design matrix.

The main function in this package is semiauto_fit(). This function fits the PLM-AR model, picks the AR order, applies the Group SCAD penalty, and stores important outputs such as fitted values and residuals. It also gives warnings when predictors are highly correlated or when a variable has too few unique values.

Finally, semiauto_plot() displays the model results through diagnostic plots, including nonlinear curves, residuals, residual autocorrelation, observed versus fitted values, structure identification, and AR coefficients.

4. Empirical Application

4.1 Data: Turkish Macroeconomic Indicators

We use macro_turkey, a monthly dataset containing 133 observations of Turkish macroeconomic indicators from January 2015 to January 2026. The response variable is the monthly inflation rate (inflasi). The five predictor variables used in the model are shown in Table 3.

Table 3. Variables in the macro_turkey dataset

Variable	Description
inflasi	Monthly inflation rate (response variable)
kurs	USD/TRY exchange rate
m3	Broad money supply (M3)
industri	Industrial production index
kredit	Total credit volume
brent	Brent crude oil price (USD per barrel)

Because the observations are collected monthly over time, the error terms may be autocorrelated. Therefore, this dataset is suitable for illustrating the PLM-AR approach used in semiautoReg. The main empirical question is whether each predictor has a linear effect on inflation or whether it should be modelled using a more flexible nonlinear function.

4.2 Workflow

The main steps of an analysis using `semiautoReg` are shown in Figure 1. The workflow begins with data preparation, continues with model fitting, and ends with model inspection and diagnostic visualization.

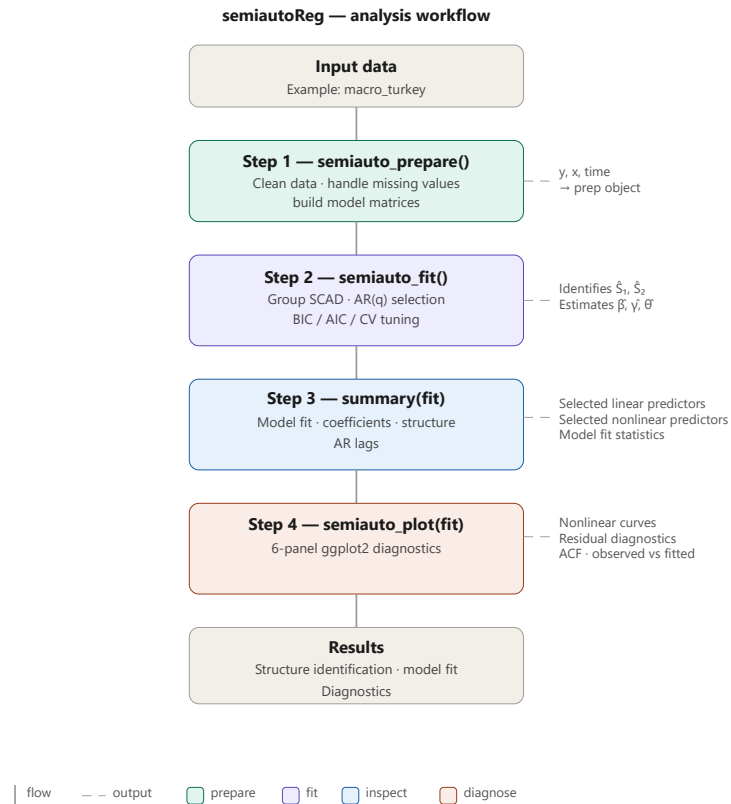


Figure 1. General workflow of the semiautoReg package

4.3 Results

The algorithm identified kurs (exchange rate), m3 (money supply), kredit, and brent as variables with nonlinear effects on inflation, while industri was retained in the linear component. However, the model also gave a diagnostic warning because kurs and m3 have a very high correlation ($|\text{cor}| = 0.992$). Because of this, the nonlinear effects of these two variables should be interpreted carefully, since structure identification may become unstable when predictors are strongly correlated.

The tuning parameters were selected using BIC, with $\lambda = 1.9496$ and $\mu = 1$. The selected AR order was $q = 4$, but AR(4) was shrunk to zero after penalization. Therefore, the active AR structure consists of AR(1), AR(2), and AR(3). The model was fitted with $\text{max_iter} = 1000$, and convergence was reached after 991 iterations.

The model fit was strong, with $R^2 = 0.9845$, $\text{RMSE} = 2.851$, and residual sum of squares = 1081.073. The estimated nonlinear functions support the structure identification result, showing that kurs, m3, kredit, and brent are better represented as nonlinear components rather than simple linear terms.

The residual diagnostics indicate that the AR correction helps reduce serial dependence. Overall, the results suggest that the fitted PLM-AR model provides a good description of the macroeconomic data, while the high correlation between kurs and m3 should still be kept in mind when interpreting the individual effects.

The main diagnostic plots produced by `semiauto_plot()` are presented in Figures 2–4. Figure 2 presents the structure identification results. Figure 3 displays the estimated nonlinear component functions for the variables classified as nonlinear. Figure 4 compares the observed and fitted inflation series.

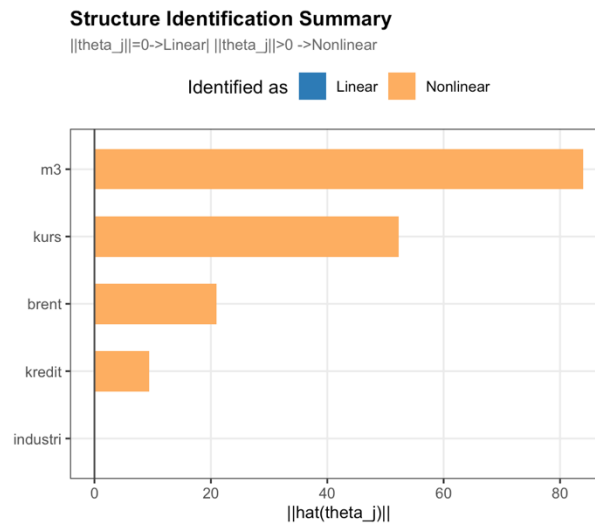


Figure 2. Structure Identification Summary

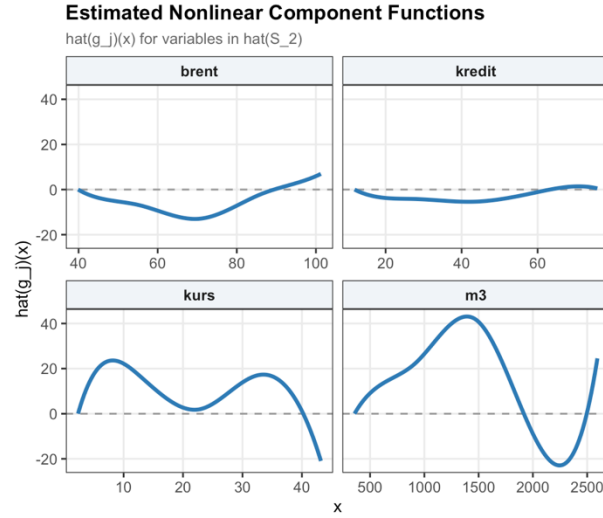


Figure 3. Estimated Nonlinear Component Functions

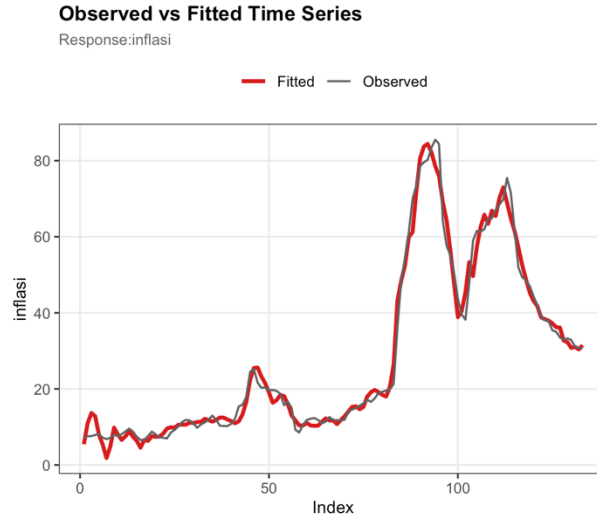


Figure 4. Observed vs Fitted Time Series

5. Computational Details

The computational cost mainly comes from the two-dimensional grid search used to select the tuning parameters λ and μ . Under the default settings, the package evaluates 20 values of λ and 15 values of μ . For each combination, the model is estimated using the BCD algorithm. As a result, BIC and AIC tuning are generally fast, while cross-validation requires more computation because the model must be refitted for each fold.

Several measures are included to improve numerical stability. The spline coefficients are initialized using a ridge-regularized least squares solution, which provides a more stable starting point for the optimization algorithm. In addition, a Moore–Penrose pseudoinverse based on `MASS::ginv()` is used when the normal equations become nearly singular.

The package also provides warning messages for common data issues. These include high correlation between predictors, a small number of unique values relative to the spline basis

dimension, and limited sample size compared with the complexity of the fitted model. Such warnings help users assess the reliability of the identified model structure and estimation results.

6. Summary and Conclusions

This paper introduced semiautoReg, an R package for structure identification in partially linear regression models with autoregressive errors. The package is based on the penalized identification method of Kazemi et al. (2021). It provides a simple workflow for checking data, preparing the model, fitting the PLM-AR model, and producing diagnostic plots.

The application using the macro_turkey dataset showed that semiautoReg can separate linear and nonlinear predictor effects automatically. In this example, industri was identified as a linear predictor, while kurs, m3, kredit, and brent were identified as nonlinear predictors. The model also included an AR error structure to account for serial dependence in the monthly data.

Overall, semiautoReg offers a practical tool for users who want to fit partially linear models when the model structure is not known in advance. The current version is limited to continuous responses and AR-type error structures, and structure identification should be interpreted carefully when predictors are highly correlated. Future development may include support for MA or ARMA error structures, faster grid search through parallel computation, and extensions to non-Gaussian response variables.

7. Acknowledgements

The author would like to thank the supervisors and colleagues at Muğla Sıtkı Koçman University for their comments and suggestions during the development of this package. The macro_turkey dataset was compiled from publicly available sources from the Central Bank of the Republic of Türkiye and TurkStat.

8. References

- Aneiros, G., & Vieu, P. (2013). Testing linearity in semi-parametric functional data analysis. *Computational Statistics*, 28(2), 413–434.
- Breheny, P., & Huang, J. (2015). Group descent algorithms for nonconvex penalized linear and logistic regression models with grouped predictors. *Statistics and Computing*, 25(2), 173–187.
- Brockwell, P. J., & Davis, R. A. (1991). *Time Series: Theory and Methods* (2nd ed.). Springer-Verlag.
- de Boor, C. (2001). *A Practical Guide to Splines* (Rev. ed.). Springer.
- Fan, J., & Li, R. (2001). Variable selection via nonconcave penalized likelihood and its oracle properties. *Journal of the American Statistical Association*, 96(456), 1348–1360.
- Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.
- Kazemi, M., Shahsvani, D., Arashi, M., & Rodrigues, P. C. (2021). Identification for partially linear regression model with autoregressive errors. *Journal of Statistical Computation and Simulation*, 91(7), 1441–1454.
- Stasinopoulos, D. M., & Rigby, R. A. (2007). Generalized additive models for location, scale and shape (GAMLSS) in R. *Journal of Statistical Software*, 23(7), 1–46.
- Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R* (2nd ed.). Chapman and Hall/CRC.

You, J., & Chen, G. (2007). On inference for a semiparametric partially linear regression model with serially correlated errors. *The Canadian Journal of Statistics*, 35(4), 515–531.